

ETD- θ : Removing the ε -Dependent Quadrature Bias from the ETD Fokker–Planck Projection

Optimal Transport / Schrödinger Bridge Project

Implements `projection_expsemi_theta.py`; reference scheme is `projection_expsemi.py`

Abstract

The ETD projection in `projection_expsemi.py` integrates the diffusion term *exactly*, yet its accuracy still degrades as ε grows: measured on the FP solution error, its order falls from 2.00 at $\varepsilon = 10^{-2}$ to 1.28 at $\varepsilon = 10$, losing a factor 60 in absolute accuracy. This note shows the cause is not the semigroup treatment but a *source-term quadrature bias*: ETD1 samples $D_x m$ at the time midpoint, while the exponential weight it is integrated against has its centroid at $\theta(\alpha) > 1/2$. Correcting the sample point — by interpolating between the two m rows the staggered grid already provides, with weight $\omega = \theta(\alpha) - 1/2$ — restores second order uniformly in ε with a constant flat across four decades, at the cost of one extra precomputed array. The correction preserves the tridiagonal-in- t structure, so the existing batched Thomas sweep still applies, and it degenerates to ETD1 identically at $\omega = 0$. Sections 5 and 10 record two things that did *not* go as expected: the final time row forces a specific extrapolation choice (a linear one destroys the band structure), and the end-to-end LADMM confirmation at large ε is currently inconclusive for reasons unrelated to the scheme.

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1 Setup and notation

Throughout, the FP constraint is

$$\partial_t \rho + \partial_x m = \varepsilon \partial_{xx} \rho, \quad (1)$$

discretised on the staggered grid of `operators.py`: ρ at $(n_t - 1) \times n_x$ (time-interior nodes, space cell-centres), m at $n_t \times (n_x - 1)$ (time cell-centres, space-interior nodes), and the residual on the ϕ grid $n_t \times n_x$.

Every scheme here is diagonal in the DCT- x basis. Writing λ_j for the discrete Neumann Laplacian eigenvalues of `grid.py`,

$$\lambda_j = \frac{2 - 2 \cos(j\pi/n_x)}{\Delta x^2}, \quad \alpha_j := \varepsilon \lambda_j \Delta t, \quad (2)$$

and hatting DCT- x coefficients, ETD1 reads, per mode j and time row $k = 0, \dots, n_t - 1$,

$$\frac{\hat{\rho}_{k+1} - c_j \hat{\rho}_k}{\Delta t} + \varphi_j \hat{g}_k = 0, \quad c_j = e^{-\alpha_j}, \quad \varphi_j = \frac{1 - c_j}{\alpha_j}, \quad (3)$$

where $g := D_x m$ and D_x is `deriv_x_at_phi`. The boundary rows use $\hat{\rho}_0 = \hat{\rho}_0^{\text{BC}}$ and $\hat{\rho}_{n_t} = \hat{\rho}_1^{\text{BC}}$. Note $\hat{g}_k$ is m at the time *midpoint* $t_{k+1/2}$, since that is where the staggered grid places m .

2 The error is a quadrature bias, not a semigroup error

The factor $c_j = e^{-\alpha_j}$ in (3) is the *exact* semigroup factor for the space-discrete problem, so the homogeneous part of ETD1 carries no error at any ε . This is worth stating sharply because it is verifiable: feeding ETD1 the semi-discrete pure-diffusion solution $\rho(t, x_i) = 1 + a \cos(n\pi x_i) e^{-\varepsilon \lambda_n t}$ with $m \equiv 0$ produces a residual of 10^{-14} for every ε from 10^{-3} to 10 , whereas Crank–Nicolson on the same data produces 1.6×10^{-9} up to 1.2×10^1 over that range.

The error lives entirely in the source term. Integrating (1) exactly over one step gives

$$\hat{\rho}_{k+1} = c_j \hat{\rho}_k - \int_{t_k}^{t_k + \Delta t} e^{-\varepsilon \lambda_j (t_{k+1} - s)} \hat{g}(s) ds. \quad (4)$$

ETD1 approximates the integral by $\Delta t \varphi_j \hat{g}(t_{k+1/2})$. The weight *integral* is exact, since $\int_{t_k}^{t_{k+1}} e^{-\varepsilon \lambda_j (t_{k+1} - s)} ds = \Delta t \varphi_j$. What is wrong is the sample point: the weight

$$w(s) = e^{-\varepsilon \lambda_j (t_{k+1} - s)} \quad (5)$$

is not symmetric about the midpoint — it increases monotonically towards $s = t_{k+1}$. Expanding $\hat{g}$ about the midpoint $t_m := t_{k+1/2}$,

$$\int w(s) [\hat{g}(s) - \hat{g}(t_m)] ds = \hat{g}'(t_m) \int w(s) (s - t_m) ds + O(\Delta t^3), \quad (6)$$

and the first-order term survives precisely because t_m is not the weight's centroid.

Fact 1 (the bias). *With $\theta(\alpha)$ the w -weighted centroid of $[t_k, t_{k+1}]$ expressed as a fraction of Δt , the leading ETD1 error per step is*

$$\Delta t \varphi_j \cdot \Delta t \left(\theta(\alpha_j) - \frac{1}{2} \right) \hat{g}'(t_m). \quad (7)$$

It vanishes at $\varepsilon = 0$ (flat weight, centroid = midpoint) and saturates as $\alpha \rightarrow \infty$. This is the ε -dependent error constant.

3 The centroid weight $\theta(\alpha)$

Substitute $u = (s - t_k)/\Delta t$, so $u \in [0, 1]$ and $w(u) = e^{-\alpha(1-u)}$. The two moments are

$$\int_0^1 e^{-\alpha(1-u)} du = \frac{1 - e^{-\alpha}}{\alpha} =: \varphi(\alpha), \quad (8)$$

$$\int_0^1 u e^{-\alpha(1-u)} du = \frac{(\alpha - 1) + e^{-\alpha}}{\alpha^2} =: \psi(\alpha), \quad (9)$$

the second by parts. Their ratio is the centroid:

$$\boxed{\theta(\alpha) = \frac{\psi(\alpha)}{\varphi(\alpha)} = \frac{(\alpha - 1) + e^{-\alpha}}{\alpha(1 - e^{-\alpha})}} \quad (10)$$

Limits. Expanding (8)–(9), $\psi = \frac{1}{2} - \frac{\alpha}{6} + \frac{\alpha^2}{24} - \frac{\alpha^3}{120} + \dots$ and $1/\varphi = 1 + \frac{\alpha}{2} + \frac{\alpha^2}{12} + 0 \cdot \alpha^3 + \dots$, so

$$\theta(\alpha) = \frac{1}{2} + \frac{\alpha}{12} - \frac{\alpha^3}{720} + O(\alpha^5), \quad (11)$$

with the α^2 and α^4 terms vanishing. As $\alpha \rightarrow \infty$, $e^{-\alpha} \rightarrow 0$ and $\theta \rightarrow (\alpha - 1)/\alpha = 1 - 1/\alpha \rightarrow 1$: the weight collapses onto the right endpoint. Hence $\theta \in [\frac{1}{2}, 1)$, monotone increasing, and the midpoint is correct only to $O(\alpha)$.

The interpolation weight. Define

$$\omega(\alpha) := \theta(\alpha) - \frac{1}{2} \in [0, \frac{1}{2}). \quad (12)$$

The target sample point $t_k + \theta\Delta t$ lies in $[t_{k+1/2}, t_{k+1}]$. The two nearest m nodes are $t_{k+1/2}$ and $t_{k+3/2}$, and the target sits at fractional position $\theta + \frac{1}{2} \in [1, 1.5]$ between them — always *inside* that pair, never an extrapolation. Linear interpolation therefore gives

$$\hat{g}(t_k + \theta\Delta t) \approx (1 - \omega) \hat{g}_k + \omega \hat{g}_{k+1}. \quad (13)$$

The first-order bias (7) is removed by construction; what remains is the $O(\Delta t^2)$ interpolation error times $\int w = \Delta t \varphi$, and since $\varphi \leq 1$ for all α , the remaining constant is bounded uniformly in ε .

Lemma 1 (relation to ETD2). (13) is ETD2's quadrature. ETD2 integrates the linear interpolant of $\hat{g}$ through two integer-node values exactly against w , giving $\varphi[(1 - \theta)\hat{g}(t_k) + \theta\hat{g}(t_{k+1})]$ with $\theta = \psi/\varphi$ as in (10) — the same rule, evaluated on a different grid.

The practical consequence is that the $n_t + 1$ integer-node relayout assumed by the (unimplemented) `cfg_ladmm_gaussian_expsemi_etd2.m` is unnecessary: the half-integer nodes already carry enough information to achieve ETD2's order.

4 The scheme and its normal equations

ETD- θ replaces (3) by

$$[R(\rho, m)]_k := \frac{\hat{\rho}_{k+1} - c_j \hat{\rho}_k}{\Delta t} + \varphi_j [\Theta \hat{g}]_k = 0, \quad (14)$$

with Θ the time-only operator implementing (13). Only the m block changes; the ρ block is bit-for-bit ETD1's.

4.1 Θ and its adjoint

As implemented (`_apply_theta`, `_apply_theta_adj`), with ω per-mode:

$$[\Theta v]_k = \begin{cases} (1 - \omega)v_k + \omega v_{k+1}, & 0 \leq k \leq n_t - 2, \\ v_{n_t-1}, & k = n_t - 1, \end{cases} \quad (15)$$

so as a matrix Θ is upper bidiagonal with diagonal $(1 - \omega, \dots, 1 - \omega, 1)$. Accumulating $[\Theta^* v]_l = \sum_k \Theta_{kl} v_k$ column by column,

$$[\Theta^* v]_l = \begin{cases} (1 - \omega)v_0, & l = 0, \\ (1 - \omega)v_l + \omega v_{l-1}, & 1 \leq l \leq n_t - 2, \\ v_{n_t-1} + \omega v_{n_t-2}, & l = n_t - 1. \end{cases} \quad (16)$$

4.2 The per-mode Gram matrix

Since Θ acts on time and D_x on space, they commute, and $R_m = \varphi \Theta D_x$ gives $R_m R_m^* = \varphi^2 \Theta (D_x D_x^*) \Theta^* = \varphi^2 \lambda \Theta \Theta^*$. With $R_\rho R_\rho^*$ unchanged from ETD1, the per-mode system is

$$T_j = R_\rho R_\rho^* + \varphi_j^2 \lambda_j \Theta \Theta^*. \quad (17)$$

Both terms are tridiagonal, hence so is T_j . Writing $\Gamma := \varphi_j^2 \lambda_j$, the entries implemented in `precomp_expsemi_theta_proj` are

row	main diagonal	off-diagonal (row k , $k+1$)
$k = 0$	$1/\Delta t^2 + \Gamma[(1 - \omega)^2 + \omega^2]$	$-c/\Delta t^2 + \Gamma \omega(1 - \omega)$
$1 \leq k \leq n_t - 3$	$(1 + c^2)/\Delta t^2 + \Gamma[(1 - \omega)^2 + \omega^2]$	$-c/\Delta t^2 + \Gamma \omega(1 - \omega)$
$k = n_t - 2$	$(1 + c^2)/\Delta t^2 + \Gamma[(1 - \omega)^2 + \omega^2]$	$-c/\Delta t^2 + \Gamma \omega$
$k = n_t - 1$	$c^2/\Delta t^2 + \Gamma$	—

The two irregular entries both come from Θ 's final row being a lone 1: its Gram diagonal is 1 rather than $(1 - \omega)^2 + \omega^2$, and $(\Theta \Theta^*)_{n_t-2, n_t-1} = \omega$ rather than $\omega(1 - \omega)$. Setting $\omega = 0$ collapses the table to $\text{diag} = (1 + c^2)/\Delta t^2 + \varphi^2 \lambda$ with constant off-diagonal $-c/\Delta t^2$, which is exactly `precomp_expsemi_proj`: the degeneracy to ETD1 is structural, not approximate.

Because the off-diagonal is row-varying in one place, the module uses `thomas_batch_precomp_general` / `thomas_batch_solve_general` from `projection.nu.py` rather than `projection.py`'s constant-off-diagonal pair. These reduce to the constant version when `sub` is uniform, so nothing is lost.

4.3 The correction step

The projection is $x_{\text{out}} = x_{\text{in}} - R^* T^{-1} R x_{\text{in}}$. Since R_ρ is untouched, the ρ correction is ETD1's verbatim. For m , $R_m^* = D_x^* \Theta^* \varphi$, so Θ^* is applied in DCT space (where ω is diagonal) before the inverse transform:

```
p_weighted = idct(_apply_theta_adj(phi_vals * p_hat, omega), axis=1)
mx_out      = psi + ops.deriv_x_at_m(p_weighted)
```

The absent minus sign is correct: `deriv_x_at_m` is the *negated* adjoint of `deriv_x_at_phi`, as a two-point difference must be, and ETD1 relies on the same cancellation.

The DC mode. At $j = 0$, $\lambda_0 = 0$ so $\alpha_0 = 0$, giving $c = 1$, $\varphi = 1$, $\omega = 0$, and the entire m block drops out of (17). T_0 is therefore ETD1's singular Neumann-in- t Laplacian, handled by the same DCT-in- t gauge fix, unchanged.

5 The final time row forces the extrapolation choice

Row $k = n_t - 1$ of (13) needs $\hat{g}_{n_t}$, which does not exist: m has n_t rows, the last at $t = 1 - \Delta t/2$. The obvious choice is linear extrapolation, $\hat{g}_{n_t} := 2\hat{g}_{n_t-1} - \hat{g}_{n_t-2}$, which keeps that row second-order consistent. **It must not be used.**

With linear extrapolation, Θ 's final row spans columns $\{n_t - 2, n_t - 1\}$. That row then overlaps row $n_t - 3$, which spans $\{n_t - 3, n_t - 2\}$, producing

$$(\Theta\Theta^*)_{n_t-3, n_t-1} = \omega \cdot (-\omega) = -\omega^2 \neq 0, \quad (18)$$

a distance-2 entry. $\Theta\Theta^*$ becomes bandwidth 2, (17) is no longer tridiagonal, and the Thomas sweep silently solves the wrong system. This was not caught by hand — it was caught by check #5 of `scripts/validate_expsemi_theta.py`, which reported post-projection residual ratios of 3×10^{-2} instead of 10^{-15} , together with a 5×10^{-3} idempotence drift.

The argument generalises: *any* two-point final row touching column $n_t - 2$ overlaps row $n_t - 3$. A tridiagonal RR^* therefore requires the final row to touch column $n_t - 1$ alone, i.e. constant extrapolation $\hat{g}_{n_t} := \hat{g}_{n_t-1}$, which collapses that row to the plain midpoint sample — (15).

Two consolations. First, Θ is then upper triangular with $\det \Theta = (1 - \omega)^{n_t-1} \neq 0$ for $\omega < 1$, so $\Theta\Theta^*$ is SPD by inspection, with no continuity argument needed. Second, the accuracy cost is nil: Table 1 compares the two rules and they agree to every printed digit. The reason is structural — the final residual row determines $\hat{\rho}_{n_t}$, which is a prescribed boundary condition rather than an unknown.

6 Evaluating θ in floating point

The direct form (10) is catastrophically cancelling at small α : its numerator $(\alpha - 1) + e^{-\alpha} \sim \alpha^2/2$ is a difference of $O(1)$ quantities. The damage is invisible in θ and severe in ω , because $\theta \approx 1/2$ dominates its own relative error while $\omega = \theta - 1/2$ does not. At $\alpha = 10^{-4}$ the direct form is already wrong in the fourth significant digit of ω (relative error 1.5×10^{-4}) while θ still looks accurate to 10^{-12} .

Table 1: Last-row rule, relative L^2 FP solution error, rough $\sigma = 0.05$ data, $n_t = n_x = n$. “const” is what ships; “linear” is the bandwidth-2 variant.

ε	rule	$n=32$	$n=64$	$n=128$	$n=256$	$n=512$
0.1	midpoint (ETD1)	1.818e-2	4.755e-3	1.212e-3	3.046e-4	7.624e-5
	const	1.098e-2	2.780e-3	7.074e-4	1.789e-4	4.499e-5
	linear	1.098e-2	2.780e-3	7.074e-4	1.789e-4	4.499e-5
1	midpoint (ETD1)	4.399e-2	1.661e-2	5.789e-3	1.811e-3	5.100e-4
	const	9.580e-3	2.403e-3	6.262e-4	1.652e-4	4.327e-5
	linear	9.580e-3	2.403e-3	6.262e-4	1.652e-4	4.327e-5
10	midpoint (ETD1)	6.512e-2	2.967e-2	1.320e-2	5.614e-3	2.234e-3
	const	9.531e-3	2.318e-3	5.752e-4	1.444e-4	3.687e-5
	linear	9.531e-3	2.318e-3	5.752e-4	1.444e-4	3.687e-5

Both branches were compared against adaptive quadrature of (8)–(9) over $\alpha \in [10^{-6}, 3]$. They cross near $\alpha = 0.02$, where each holds about 10^{-10} relative error on ω ; the module switches there, using the two-term series (11) below and the direct form above. For large α , $e^{-\alpha}$ underflows harmlessly and (10) is well conditioned.

The lesson generalises to the validation: **check the error on ω , not on θ** . The first version of check #1 measured θ and passed a cutoff that was two orders of magnitude too low.

7 Files

file	status	what
<code>optimal_transport/projection_expsemi_theta.py</code>	new	the scheme: $\theta, \omega, \Theta, \Theta^*$, precomp, project
<code>scripts/validate_expsemi_theta.py</code>	new	the six checks of §8
<code>scripts/compare_etd1_vs_theta.py</code>	new	end-to-end LADMM comparison (§10)
<code>scripts/study_utils.py</code>	edited	run_study gained precomp/projection a
<code>optimal_transport/projection_expsemi.py</code>	untouched	remains the ETD1 reference
<code>optimal_transport/projection.py</code>	untouched	remains the CN reference

The `run_study` change defaults to the ETD1 pair, so every existing caller (`grid_convergence.py`, `multi_eps_study.py`, ...) is unaffected. The new scheme plugs into the `projection/projection.state` seam that `discretize_then_optimize` already exposed, so `pipeline.py` needed no change at all.

8 Validation

`scripts/validate_expsemi_theta.py` runs six checks over $\varepsilon \in \{0, 10^{-8}, 10^{-3}, 0.1, 1, 10, 10^3\}$ at $n_t = 32$, $n_x = 24$. Nothing trusts the hand algebra of §3–§5: each piece is checked against an independently constructed reference.

1. θ, ω in range and monotone; ω matches adaptive quadrature of (8)–(9) to $< 10^{-9}$ relative; the $\alpha \rightarrow \infty$ asymptote $\theta \sim 1 - 1/\alpha$ holds to $O(\alpha^{-2})$.

2. $\langle \Theta g, v \rangle = \langle g, \Theta^* v \rangle$ to 2.7×10^{-15} , and `_apply_theta` agrees with a densely built Θ to 4.4×10^{-16} , for $\omega \in \{0, 10^{-6}, 0.05, 0.2, 0.4999\}$.
3. The closed-form (main, sub) of (17) matches a dense $R_\rho R_\rho^* + \varphi^2 \lambda \Theta \Theta^*$ built per mode to 3.4×10^{-18} relative; every per-mode T_j is verified tridiagonal and SPD (worst $\lambda_{\min}/\lambda_{\max} = 4.4 \times 10^{-10}$ over all modes and all ε).
4. At $\varepsilon = 0$, ETD- θ reproduces `proj_fokker_planck_expsemi` *exactly* (max relative difference 0).
5. **Feasibility.** The ETD- θ residual of the projected point, recomputed with a dense Θ rather than the module's own helper, drops from $O(10^3)$ to $O(10^{-12})$ — worst after/before ratio 8.2×10^{-15} across the whole ε sweep. This is the decisive check: a projection built on a T that is not the true Gram RR^* cannot land on the constraint set, so #5 would fail if *any* of (15), (16) or (17) were wrong. It is what caught the bandwidth-2 error of §5.
6. Idempotence: $P(P(x)) = P(x)$ to 0 (exactly).

9 Measured effect

The scheme-level measurement marches each scheme forward in time from $\rho(0)$ using the exact source $g = D_x m$ and compares against the exact solution — no ADMM, no projection, so what is measured is purely FP discretisation error. Data is a manufactured solution with a Gaussian-decaying spectrum at the project's own $\sigma = 0.05$, which has genuine high-frequency content (31 x -modes in 1D, 11×11 in 2D).

Solution error, not residual norm, is the right measure here: for a stiff mode the residual carries a factor $1/\Delta t$ that the inverse operator removes, so a residual-fitted order understates every scheme and does so unevenly.

Table 2: Relative L^2 FP solution error at the finest grid, with fitted order p . 1D: $n_t = n_x = 512$. 2D: $n_t = n_x = n_y = 96$, built with the project's own `operators_2d/grid_2d`.

ε	CN		ETD1		ETD- θ	
	error	p	error	p	error	p
<i>1D</i>						
0.01	4.402e-5	2.00	4.609e-5	2.00	4.163e-5	2.00
0.1	5.093e-5	2.00	7.624e-5	2.00	4.499e-5	1.99
1	5.279e-5	2.00	5.100e-4	1.75	4.327e-5	1.93
10	5.312e-5	2.02	2.234e-3	1.28	3.687e-5	1.98
<i>2D</i>						
0.01	1.641e-3	2.01	1.730e-3	2.01	1.512e-3	2.01
0.1	2.011e-3	2.03	3.310e-3	1.93	1.695e-3	1.96
1	2.066e-3	2.06	1.491e-2	1.35	1.428e-3	1.98
10	2.167e-3	2.19	2.563e-2	1.08	1.358e-3	2.02

Three readings of Table 2.

1. **ETD- θ holds $p \approx 2$ at every ε with a constant flat in ε** (1D: 4.2, 4.5, 4.3, 3.7×10^{-5} across four decades). It is the best scheme at every ε tested and never worse anywhere. Against ETD1 it gains a factor 60 at $\varepsilon = 10$ and 12 at $\varepsilon = 1$ in 1D, 19 and 10 in 2D.
2. **ETD1 degrades exactly as (7) predicts**, p falling $2.00 \rightarrow 1.75 \rightarrow 1.28$.
3. **CN's accuracy is not the problem.** Its order and constant are both stable in ε . CN's defect is elsewhere — $R_{\text{CN}}(\infty) = -1$, so at $\varepsilon = 10$ with $\sigma = 0.05$ data 99.2% of modes have $R < 0$ and one step returns $\min \rho = -3.61$ where the exact solution has $\min \rho = +0.50$. That is a positivity failure, not an accuracy failure, and ETD- θ inherits ETD1's exact semigroup so it does not share it.

Higher dimensions. The correction is purely temporal, and time is one-dimensional in every spatial dimension: with $\alpha = \varepsilon(\lambda_x + \lambda_y)\Delta t$ still a scalar per mode, θ, ω, Θ are the identical scalar functions evaluated on an (n_x, n_y) array, and the two-row m stencil keeps RR^* tridiagonal in t . It also matters *more* as dimension grows: the fraction of modes with $\alpha > 2$ at $n = 64$, $\varepsilon = 0.01$ is 30% in 1D, 66% in 2D, 87% in 3D, since $\alpha_{\max} = 4d\varepsilon n$ is linear in d and λ is a sum.

10 Open items

On the actual Gaussian SB problem the gain is a few percent, not 12–60 $\times$. This is the central caveat and it qualifies Table 2. Plugging the *reference* SB solution into each scheme's discrete FP residual — which, since the reference solves the continuous equation, measures that scheme's truncation error on the real problem — gives:

ε	n	$\ R_{\text{ETD1}}\ $	$\ R_{\text{ETD-}\theta}\ $	gain
0.001	128	1.898e-2	1.887e-2	$1.01\times$
0.01	128	3.622e-2	3.550e-2	$1.02\times$
0.1	128	6.481e-1	6.129e-1	$1.06\times$
1	128	1.001e1	1.002e1	$1.00\times$

The reason is structural, and it is the honest limit of this work. ETD1 loses *order* only when the solution carries significant content in modes with $\alpha = \varepsilon\lambda\Delta t \gtrsim 1$; for modes it resolves, ETD1 and ETD- θ are both second order and differ by a bounded constant. The manufactured solutions of Table 2 were built with a $\sigma = 0.05$ spectrum across 31 x -modes *and* a momentum field $m \propto \varepsilon k^2 g$, so their $D_x m$ is large and fast-varying exactly in the stiff modes the correction targets. The Gaussian SB solution is not like that: it is smooth, and at large ε it becomes *smoother* (more nearly uniform), so it has almost no content where $\alpha \gtrsim 1$.

So ETD- θ removes a real, correctly identified ε -dependent error term, but that term is not the dominant one for this problem. The original question — why the error grows with ε here — remains open. Note in particular that at $\varepsilon = 0.1$ the fitted order degrades to ≈ 1 for *both* schemes equally, which rules the quadrature bias out as the cause and points at a separate ε -dependent term not yet identified (candidates: the spatial treatment of $\varepsilon\partial_{xx}$, or the reference's own boundary mismatch).

The reference is degenerate at large ε . At $\varepsilon = 1$ the residual above does not decrease under refinement at all ($8.79 \rightarrow 9.97 \rightarrow 10.01$ for $n = 32, 64, 128$): the reference does not satisfy the discrete FP equation, so nothing measured against it at that ε means anything. `sinkhorn_fit` applies the heat semigroup over the *whole* interval $T = 1$, and $e^{-\varepsilon\pi^2} = 5.2 \times 10^{-5}$ at $\varepsilon = 1$ annihilates every non-constant mode, so the iteration reaches a trivial fixed point in 2 iterations. This is adjacent to the known `SINKHORN_VAREPS_THRESHOLD` issue and needs fixing before any large- ε claim, for any scheme, can be made.

The end-to-end LADMM comparison is correspondingly uninformative. `scripts/compare_etd1_vs_t` shows ETD- θ ahead by $1.01\text{--}1.08\times$, consistent with the residual table above, with absolute errors of $O(1)\text{--}O(10)$ at $\varepsilon \geq 1$ against the degenerate reference. At $\varepsilon = 0.1$, where the reference is most trustworthy, the gain does grow monotonically with resolution ($1.03 \rightarrow 1.05 \rightarrow 1.08$) — the right direction, but a weak signal.

Where ETD- θ should pay off. A problem retaining genuine fine structure at large ε — bimodal or rough marginals, cf. `matlab/discretize_first/2d/experiments/test_expsemi_bimodal.m` — puts solution content where $\alpha \gtrsim 1$ and is the natural test. This has not been run.

Also not done.

- The `_nu` (non-uniform time grid) variant. It matters more there, since grading shrinks Δt and raises the local α . `projection_expsemi_nu.py` builds its system by numerical probing rather than hand algebra, so the correction should be absorbed with no new derivation.
- The 2D/3D port. Python 2D currently has only the banded/CN projection, so this means porting ETD anyway; it should be ported directly as ETD- θ , with `matlab/shared/2d/projection/proj_for` as the structural reference. One caution: do not materialise the lookahead $\hat{g}_{k+1}$ as a full (n_t, n_x, n_y) array — fuse it into the existing t -sweep.
- An adjoint check of the full R/R^* pair against the LADMM’s own norms. Checks #2 and #5 cover Θ and the Gram, but the assembled operator is only checked implicitly through feasibility.

Worth an independent look. Equations (9) (integration by parts), (11) (the vanishing α^2 term, which is what makes the two-term series worth using), (16) (the adjoint’s boundary rows), and the $k = n_t - 2$ off-diagonal $\Gamma\omega$ in the table of §4.2.